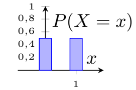
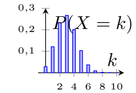
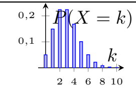
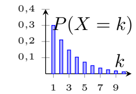
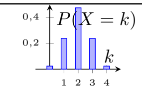
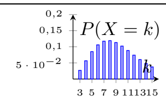
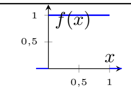
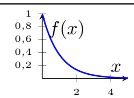
Tabla de distribuciones de probabilidad

Nombre	**Notación (parámetros)**	**Función de masa/densidad**	**Función de distribución**	**Función generadora de momentos**	**Esperanza y varianza**	**Gráfica**
Bernoulli	$X \sim \mathrm{Bernoulli}(p)$, $0 \leq p \leq 1$	$P(X=1)=p$, $P(X=0)=1-p$	$F(x) = \begin{cases} 0, & x<0 \\ 1-p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$	$M_X(t) = 1-p+pe^t$	$E[X]=p$, $\mathrm{Var}(X) = p(1-p)$	
Binomial	$X \sim \mathrm{Bin}(n,p)$, $n \in \mathbb{N}$, $0 \leq p \leq 1$	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k=0,1,\ldots,n$	$F(x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}$	$M_X(t) = (1-p+pe^t)^n$	$E[X]=np$, $\mathrm{Var}(X) = np(1-p)$	
Poisson	$X \sim \mathrm{Poisson}(\lambda)$, $\lambda > 0$	$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $k=0,1,2,\ldots$	$F(x) = e^{-\lambda} \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k}{k!}$	$M_X(t) = \exp\{\lambda(e^t-1)\}$	$E[X]=\lambda$, $\mathrm{Var}(X)=\lambda$	
Geométrica	$X \sim \mathrm{Geom}(p)$, $0 < p \leq 1$, $x=1,2,\ldots$	$P(X=k) = (1-p)^{k-1}p$, $k=1,2,\ldots$	$F(x) = 1-(1-p)^{\lfloor x \rfloor}$	$M_X(t) = \frac{pe^t}{1-(1-p)e^t}$, para $t < -\ln(1-p)$	$E[X]=\frac{1}{p}$, $\mathrm{Var}(X) = \frac{1-p}{p^2}$	
Hipergeométrica	$X \sim$ Hipergeométrica (N,K,n), $N \in \mathbb{N}$, $K \in \{0,\ldots,N\}$, $n \in \{0,\ldots,N\}$	$P(X=k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$, $\max(0,n-(N-K)) \leq k \leq \min(n,K)$	$F(x) = \sum_{k=0}^{\lfloor x \rfloor} \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	No disponible en forma cerrada simple	$E[X]=n\frac{K}{N}$, $\mathrm{Var}(X) = n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$	
Binomial Negativa	$X \sim \mathrm{NB}(r,p)$, $r>0$, $0<p\leq 1$, $x=r,r+1,\ldots$	$P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$, $k=r,r+1,\ldots$	$F(x) = \sum_{k=r}^{\lfloor x \rfloor} \binom{k-1}{r-1} p^r (1-p)^{k-r}$	$M_X(t) = \left(\frac{pe^t}{1-(1-p)e^t}\right)^r$, $t < -\ln(1-p)$	$E[X]=\frac{r}{p}$, $\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$	
Uniforme	$X \sim \mathcal{U}(a,b)$, $a<b$	$f(x) = \frac{1}{b-a}$, $a \leq x \leq b$	$F(x) = \begin{cases} 0, & x<a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x>b \end{cases}$	$M_X(t) = \frac{e^{tb}-e^{ta}}{t(b-a)}$, $t \neq 0$	$E[X]=\frac{a+b}{2}$, $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$	
Exponencial	$X \sim \mathrm{Exp}(\lambda)$, $\lambda>0$	$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$	$F(x) = 1-e^{-\lambda x}$, $x \geq 0$	$M_X(t) = \frac{\lambda}{\lambda - t}$, $t<\lambda$	$E[X]=\frac{1}{\lambda}$, $\mathrm{Var}(X) = \frac{1}{\lambda^2}$	

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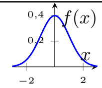
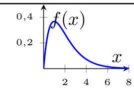
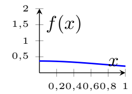
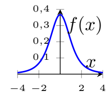
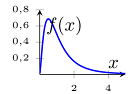
Nombre	Notación (parámetros)	Función de masa/densidad	Función de distribución	Función generadora de momentos	Esperanza y varianza	Gráfica
Normal	$X\sim\mathcal{N}(\mu,\sigma^2)$, $\mu\in\mathbb{R}$, $\sigma>0$	$f(x)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$, $x\in\mathbb{R}$	$F(x)=\Phi\left(\frac{x-\mu}{\sigma}\right)$, donde Φ es la CDF normal estándar	$M_X(t)=\exp\left\{\mu t+\frac{1}{2}\sigma^2t^2\right\}$	$E[X]=\mu$, $\mathrm{Var}(X)=\sigma^2$	
Gamma	$X\sim\mathrm{Gamma}(\alpha,\beta)$, $\alpha>0$ (forma), $\beta>0$ (tasa)	$f(x)=\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$, $x>0$	$F(x)=\frac{\gamma(\alpha,\beta x)}{\Gamma(\alpha)}$, donde γ es la función gamma incompleta	$M_X(t)=\left(\frac{\beta}{\beta-t}\right)^\alpha$, $t<\beta$	$E[X]=\frac{\alpha}{\beta}$, $\mathrm{Var}(X)=\frac{\alpha}{\beta^2}$	
Beta	$X\sim\mathrm{Beta}(\alpha,\beta)$, $\alpha>0$, $\beta>0$	$f(x)=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$, $0<x<1$	$F(x)=I_x(\alpha,\beta)$ (función beta incompleta regularizada)	$M_X(t)=1+\sum_{k=1}^{\infty}\left(\prod_{r=0}^{k-1}\frac{\alpha+r}{\alpha+\beta+r}\right)\frac{t^k}{k!}$	$E[X]=\frac{\alpha}{\alpha+\beta}$, $\mathrm{Var}(X)=\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
t-Student	$X\sim t_\nu$, $\nu>0$ (grados de libertad)	$f(x)=\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$, $x\in\mathbb{R}$	$F(x)=\frac{1}{2}+\frac{x\,\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\,{}_2F_1\left(\frac{1}{2},\frac{\nu+1}{2};\frac{3}{2};-\frac{x^2}{\nu}\right)$ (función hipergeométrica)	No existe en un entorno de cero	$E[X]=0$ para $\nu>1$; $\mathrm{Var}(X)=\frac{\nu}{\nu-2}$ para $\nu>2$	
Fisher-Snedecor	$X\sim F_{d_1,d_2}$, $d_1>0$, $d_2>0$ (grados de libertad)	$f(x)=\frac{\sqrt{\frac{(d_1x)^{d_1}d_2^{d_2}}{(d_1x+d_2)^{d_1+d_2}}}}{x\,\mathrm{B}\left(\frac{d_1}{2},\frac{d_2}{2}\right)}$, $x>0$	$F(x)=I_{\frac{d_1x}{d_1x+d_2}}\left(\frac{d_1}{2},\frac{d_2}{2}\right)$ (beta incompleta)	No existe en un entorno de cero	$E[X]=\frac{d_2}{d_2-2}$ para $d_2>2$; $\mathrm{Var}(X)=\frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$ para $d_2>4$	
Chi-cuadrado	$X\sim\chi^2_k$, $k>0$ (grados de libertad)	$f(x)=\frac{1}{2^{k/2}\Gamma(k/2)}x^{k/2-1}e^{-x/2}$, $x>0$	$F(x)=\frac{\gamma(\frac{k}{2},\frac{x}{2})}{\Gamma(\frac{k}{2})}$ (gamma incompleta)	$M_X(t)=(1-2t)^{-k/2}$, para $t<\frac{1}{2}$	$E[X]=k$, $\mathrm{Var}(X)=2k$